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Probability

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Introduction

- The term **probability** refers to the study of **randomness and uncertainty**.
- In any situation in which one of a number of possible outcomes may occur, the discipline of probability provides methods for **quantifying the chances, or likelihoods**, associated with the various outcomes.
- **Examples:**
 - There is a 50–50 chance that the incumbent will seek re-election.
 - There will probably be at least one section of that course offered next year.
 - The likelihood that tomorrow will rain is 70 percent.

2.1 Sample Spaces and Events

Sample Spaces and Events

An **experiment** is any activity or process **whose outcome is subject to uncertainty**.

Although the word *experiment* generally suggests a planned or carefully controlled laboratory testing situation, we use it here in a much wider sense.

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Sample Spaces and Events

➤ Examples:

- Tossing a coin once or several times
- Weighing a loaf of bread
- Ascertaining the commuting time from home to work on a particular morning
- Obtaining blood types from a group of individuals
- Measuring the compressive strengths of different steel beams.



The Sample Space of an Experiment

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The Sample Space of an Experiment

Definition

The **sample space** of an experiment, denoted by $\mathcal{S}$, is the **set of all possible outcomes** of that experiment.

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Example

The simplest experiment to which probability applies is one with two possible outcomes.

One such experiment consists of examining a single fuse to see whether it is defective.

- The sample space for this experiment can be abbreviated as

$\mathcal{S} = \{N, D\}$, where

N represents not defective,

D represents defective,

and the braces are used to enclose the elements of a set.

Example

cont'd

- Another such experiment would involve **tossing a thumbtack** and noting whether it landed point up or point down, with sample space

$$\mathcal{S} = \{U, D\},$$

- another would consist of observing the gender of the next child born at the local hospital, with

$$\mathcal{S} = \{M, F\}.$$

Example

cont'd

- A reasonably large percentage of C++ programs written at a particular company compile on the first run, but some do not.
- Suppose an **experiment** consists of selecting and compiling C++ programs at this location one by one **until encountering a program that compiles on the first run**.

$$\mathcal{S} = \{S, FS, FFS, FFFS, \dots\},$$

Example

cont'd

- Two gas stations are located at a certain intersection. Each one has six gas pumps. Consider the experiment in which the number of pumps in use at a particular time of day is determined for each of the stations.
- An **experimental outcome** specifies **how many pumps are in use at the first station and how many are in use at the second one.**

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Example

cont'd

		<i>Second Station</i>						
		0	1	2	3	4	5	6
<i>First Station</i>	0	(0, 0)	(0, 1)	(0, 2)	(0, 3)	(0, 4)	(0, 5)	(0, 6)
	1	(1, 0)	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2	(2, 0)	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	3	(3, 0)	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
	4	(4, 0)	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
	5	(5, 0)	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
	6	(6, 0)	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)



Events

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Events

In our study of probability, we will be interested not only in the **individual outcomes** of $\mathcal{S}$ but also in **various collections of outcomes** from $\mathcal{S}$.

Definition

An **event** is any collection (subset) of outcomes contained in the sample space $\mathcal{S}$.

An event is **simple** if it consists of **exactly one outcome** and **compound** if it consists of **more than one outcome**.

Events

- When an experiment is performed, a particular event A is said to occur if the resulting experimental outcome is contained in A .
- In general, exactly one simple event will occur, but many compound events will occur simultaneously.

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Example

Consider an experiment in which each of three vehicles taking a particular freeway exit turns left (L) or right (R) at the end of the exit ramp.

The eight possible outcomes that comprise the sample space are LLL , RLL , LRL , LLR , LRR , RLR , RRL , and RRR .

Thus there are eight simple events, among which are $E_1 = \{LLL\}$ and $E_5 = \{LRR\}$.

Example

cont'd

Some **compound events** include

the event that exactly one of the three vehicles turns right:

$$A = \{RLL, LRL, LLR\}$$

The event that at most one of the vehicles turns right:

$$B = \{LLL, RLL, LRL, LLR\}$$

The event that all three vehicles turn in the same direction:

$$C = \{LLL, RRR\}$$

Example

cont'd

Suppose that when the experiment is performed, the outcome is LLL .

Then the simple event E_1 has occurred and so also have the events B and C (but not A).

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Compound events in six-pump gas stations

$A = \{(0, 0), (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$ = the event that the number of pumps in use is the same for both stations

$B = \{(0, 4), (1, 3), (2, 2), (3, 1), (4, 0)\}$ = the event that the total number of pumps in use is four

$C = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$ = the event that at most one pump is in use at each station

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Some Relations from Set Theory

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Some Relations from Set Theory

An event is just a set, so relationships and results from elementary set theory can be used to study events.

The following operations will be used to create new events from given events.

Definition,

1. **The complement of an event** A , denoted by A' , is the set of all outcomes in $\mathcal{S}$ that are not contained in A .

Some Relations from Set Theory

2. **The union of two events** A and B , denoted by $A \cup B$ and read “ A or B ,” is the event consisting of all outcomes that are *either in A or in B or in both events* (so that the union includes outcomes for which both A and B occur as well as outcomes for which exactly one occurs) that is, all outcomes in at least one of the events.
3. **The intersection of two events** A and B , denoted by $A \cap B$ and read “ A and B ,” is the event consisting of all outcomes that are in *both A and B* .

Example 8

For the experiment in which the number of pumps in use at a single six-pump gas station is observed,

let $A = \{0, 1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{1, 3, 5\}$.

Then

$$A' = \{5, 6\},$$

$$A \cup B = \{0, 1, 2, 3, 4, 5, 6\} = \mathcal{S},$$

$$A \cup C = \{0, 1, 2, 3, 4, 5\},$$

$$A \cap B = \{3, 4\},$$

$$A \cap C = \{1, 3\},$$

$$(A \cap C)' = \{0, 2, 4, 5, 6\}$$

Some Relations from Set Theory

Sometimes A and B have no outcomes in common, so that the intersection of A and B contains no outcomes.

Definition

Let $\emptyset$ denote the *null event* (the event consisting of no outcomes whatsoever).

When $A \cap B = \emptyset$, A and B are said to be **mutually exclusive** or **disjoint** events.

Example

A small city has two automobile dealerships: a first dealer selling BRAND1 and BRAND2; a second dealer selling BRAND3 and BRAND4.

If an experiment consists of observing the brand of the next car sold, then the events

$A = \{\text{BRAND1}, \text{BRAND2}\}$ and

$B = \{\text{BRAND3}, \text{BRAND4}\}$

are **mutually exclusive** because

the next car sold cannot be both a first dealer product and a second dealer product (at least until the two companies merge!).

Some Relations from Set Theory

The operations of union and intersection can be extended to more than two events.

For any three events A , B , and C , the event $A \cup B \cup C$ is the set of outcomes **contained in at least one of the three events**, whereas $A \cap B \cap C$ is the set of outcomes **contained in all three events**.

Given events $A_1, A_2, A_3, \dots$, these events are said to be **mutually exclusive (or pairwise disjoint)** if no two events have any outcomes in common.

Some Relations from Set Theory

A pictorial representation of events and manipulations with events is obtained by using Venn diagrams.

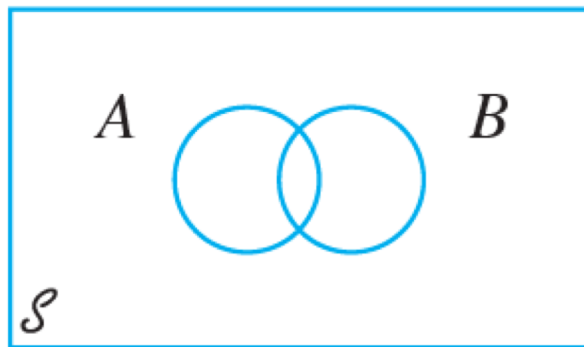
To construct a Venn diagram, draw a rectangle whose interior will represent the sample space $\mathcal{S}$.

Then any event A is represented as the interior of a closed curve (often a circle) contained in $\mathcal{S}$.

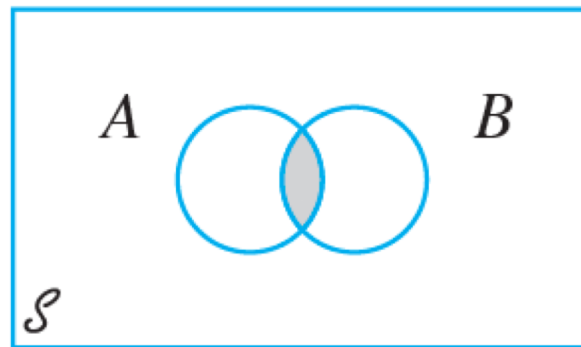
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Some Relations from Set Theory

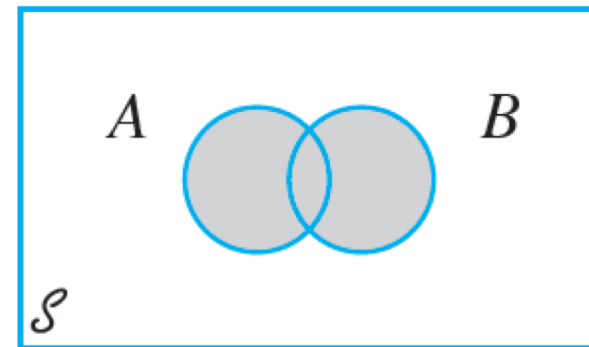
Examples of **Venn diagrams**:



(a) Venn diagram of events A and B



(b) Shaded region is $A \cap B$

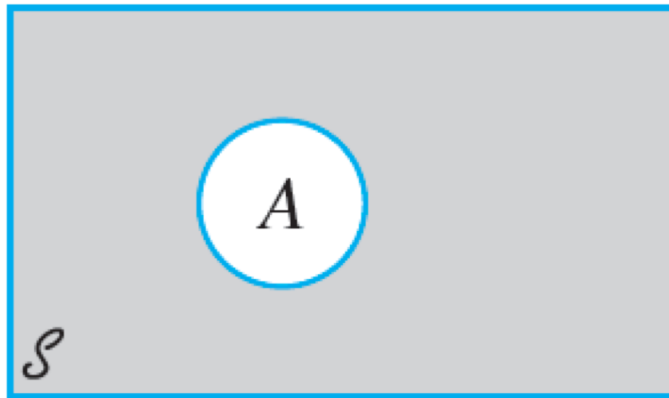


(c) Shaded region is $A \cup B$

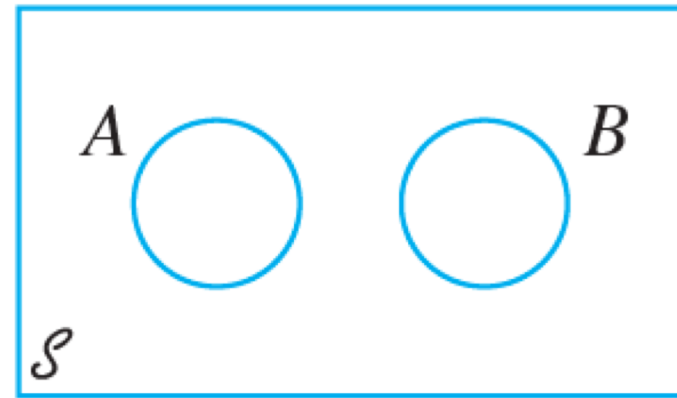
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Some Relations from Set Theory

Examples of Venn diagrams:



(d) Shaded region
is A'

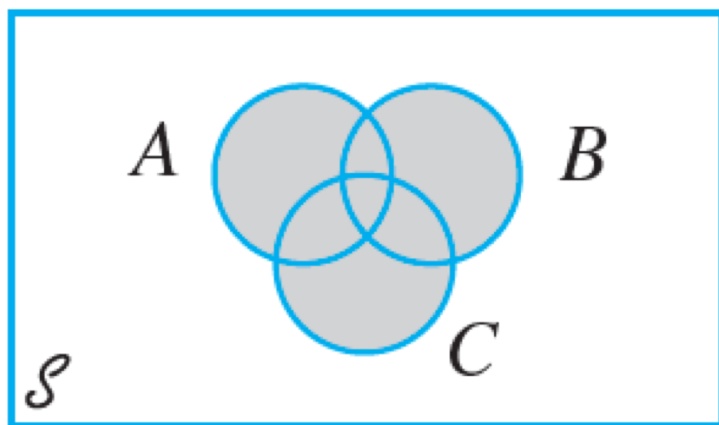


(e) Mutually exclusive
events

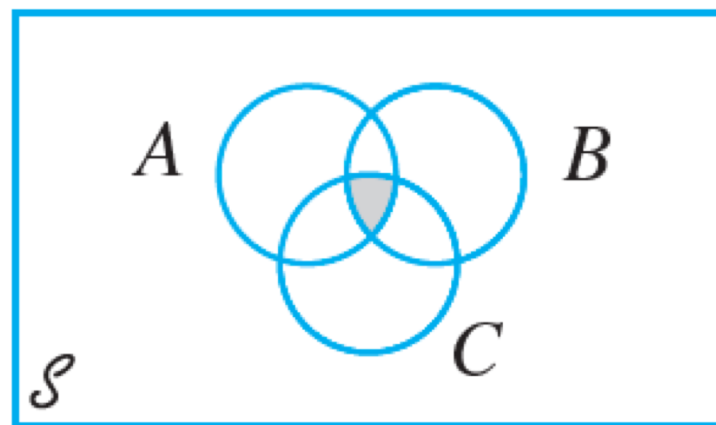
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Some Relations from Set Theory

Examples of Venn diagrams:



(f) Shaded region
is $A \cup B \cup C$



(g) Shaded region
is $A \cap B \cap C$

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